

16 CNC Software Messages

16.1 CNC Software Startup Errors and Messages

Error	Message	Cause & Effect	Action
102	Error initializing CPU... cannot continue	Error sending a hex file program to the motion control card.	Inspect MPU11 connection or fix missing/corrupted hex file. Contact Dealer.
103	Error sending setup	Error sending the current setup parameters to the motion control card.	Inspect MPU11 connection or fix missing/corrupted hex file. Contact Dealer.
104	Error sending PID setup	Error sending the current PID setup parameters to the motion control card.	Inspect MPU11 connection or fix missing/corrupted hex file. Contact Dealer.
105	mpu.plc file read error... cannot continue	Error sending the current PLC program to the motion control card.	Install or recompile the PLC program. Contact dealer.
106	The PC clock appears to be wrong	Error while reading the temperature file. The time on the PC internal clock is earlier than the time recorded in a previously-stored file.	Set the PC internal clock to the correct time.
199	CNC started	The CNC control software has started.	No action required.

16.2 Messages Issued Upon Exit from the CNC Software

Error	Message	Cause & Effect	Action
201	Exiting CNC due to a known error (ACORN only)	MPU11 is not responding, or the mpu11.hex, mpu.plc files are missing or damaged.	Check for possible software corruption. Contact dealer.
202	Exiting CNC due to a math error	A floating-point math error occurred. Possible corruption of cnc.tem, cncm.job, or cncm.wcs files.	Delete corrupted files and reboot software. Contact dealer.
204	Exiting CNC... Normal Exit	The CNC control software is shutting down normally.	No action required.
222	Autotune run	Autotune has been run.	No action required.

16.3 Messages and Prompts in the Operator Status Window Status Messages

Error	Message	Cause & Effect	Action
301	Stopped	A job has ended normally or the Operator has aborted the job.	No action required.
302	Moving...	Motors are moving while a CNC program is running.	No action required.
303	Paused...	Motion is paused while a CNC program is running (FEED HOLD).	No action required.
304	MDI...	The CNC software is running in MDI mode.	No action required.

Error	Message	Cause & Effect	Action
305	Processing. . .	The CNC software is running in a mode other than MDI.	No action required.
306	Job Finished	A normal end of the CNC program.	No action required.
307	Operator abort: job canceled	ESC or CYCLE CANCEL was pressed. Job is cancelled.	No action required.
308	Waiting for input #NN	M100 or M101 is executing. The program will continue once the specified input opens or closes.	No action required.
309	Waiting for CYCLE START button	M0, M1, M100/75, or Block Mode was executed.	Press Cycle Start
310	Waiting for output #NN	M100 or M101 is executing. The program will continue once the specified output opens or closes.	No action required.
311	Waiting for memory #NN	M100 or M101 is executing. The program will continue once the specified memory bit changes to the correct state.	No action required.
312	Waiting for PLC operation (Mnn)	The PLC program is not clearing the PLC operation in progress.	No action required.
313	Waiting for dwell time	G4 executing. The program will wait for the specified dwell time and then continue.	No action required.
314	Waiting for system #NN	M100 or M101 is executing. The program will continue once the specified PLC system variable changes to the correct state.	No action required.
315	Searching. . .	A run/search is currently in progress.	No action required.
317	Waiting for automatic tool change	mfunc6.mac is executing.	No action required.
318	Operator abort probing cancelled	ESC or CYCLE CANCEL was pressed while doing a probing move.	No action required.
319	Probing cycle cancelled	The probing cycle was cancelled.	No action required.
320	Probe stuck	The probe is stuck or hit an object when it was not expecting contact.	No action required.
322	Stall: probing cancelled	The probing was cancelled because of a stall.	No action required.
323	Stall: job cancelled	The job was cancelled because of a stall.	No action required.
324	Limit: probing cancelled	The probing was cancelled because of a limit error.	No action required.
325	Limit: job cancelled	The job was cancelled because of a limit error.	No action required.
326	Fault: probing cancelled	The probing was cancelled because of a fault.	No action required.
327	Fault: job cancelled	The job was cancelled because of a fault.	No action required.
328	Cutter comp error: job cancelled	The job was cancelled because of a cutter comp error.	No action required.
329	Invalid parameter: job cancelled	The job was cancelled because of an invalid parameter.	No action required.

Error	Message	Cause & Effect	Action
330	Canned cycle error: job cancelled	The job was cancelled because of a canned cycle error.	No action required.
332	Search Failed	A run/search was unable to find the requested G-code line.	No action required.
334	Locating position to resume job. . .	A run/search is currently locating the job continuation point in the program.	No action required.
335	Emergency Stop Released	The Emergency Stop Button has been released.	No action required.
336	Digitize cancelled	ESC or CYCLE CANCEL was pressed during digitizing.	No action required.
337	Digitize complete	A digitizing routine ran to completion.	No action required.
338	Job Cancelled	ESC or CYCLE CANCEL was pressed during the job run.	No action required.
339	Jogging. . .	An axis jog key is currently pressed and the machine is moving the corresponding axis.	No action required.
340	Limit (#___) cleared	A previously-tripped limit switch is now in the "untripped" position.	No action required.
341	Probing Cycle Finished	A probing cycle ran to completion.	No action required.
342	Waiting for motion to stop	The PC is waiting for the MPU11 to complete its motion.	No action required.
343	Waiting for stop reason reset	A PC is waiting for the MPU11 to reset the stop reason (as part of the PC/MPU11 communications handshake).	No action required.
344	Feed Rate modified due to spindle	The effective feed rate has been lowered because the spindle is spinning slower than the threshold percentage of the commanded spindle speed (the threshold percent is specified in P149).	No action required.
345	Waiting for spindle to get up to speed	Job progress is paused until the actual spindle speed reaches the threshold percentage of the commanded spindle speed (the threshold percentage is specified in P149).	No action required.
346	Waiting for spindle direction	Job progress is paused until the spindle turns in the commanded direction.	No action required.
347	Reset Cleared	The reset state has been cleared	No action required.

16.4 Abnormal Stops (Faults)

Abnormal stops are detected in the following order: PLC, servo drive, spindle drive, lube, and then ESTOP. This means that if both the servo drive and spindle drive have faulted, the servo drive fault message would appear.

Error	Message	Cause & Effect	Action
401	PLC failure detected	MPU11 stopped with PLC failure bit set. Job cancelled.	Check PLC fibers and PLC logic power.

Error	Message	Cause & Effect	Action
404	Spindle drive fault detected	MPU11 stopped with spindle drive fault bit set. Job cancelled.	Check the inverter for a fault or reset the spindle contactor OCR, then cycle the EMERGENCY STOP.
405	Lubricant level low	MPU11 stopped with low lube fault bit set. The current job will finish, but nothing will work after that.	Add lube or check the low lube switch wiring, then cycle the EMERGENCY STOP.
406	Emergency Stop detected	MPU11 stopped with no fault bits set. Job cancelled.	Release the Emergency Stop button.
407	limit (#__) tripped	MPU11 stopped with limit switch tripped. Job cancelled.	Clear the limit switch.
408	Programmed action timer expired	M103's time expired before M104 was encountered. Job cancelled.	Find out why the timer expired before the specified action was completed.
409	_Axis (__) lag	This message can be caused by insufficient torque (not enough current), or insufficient speed (not enough voltage) on a given axis.	Check power and cabling between Axis motors, drives, and controller for low voltage, noise, or shorts.

Error	Message	Cause & Effect	Action
410	_ axis position error	A position error greater than one motor encoder turn was detected on any axis. All axis motion is stopped, power to the motors is released (all servo drive commands cease), and the CNC program is aborted. The probable causes of this error are: 1. The motor is wired backwards. 2. Noise is getting into the system via the motor cables (the line integrity has been violated). 3. An encoder error has occurred.	1. Try to slow jog the motor and watch the DRO position. If the position on the DRO goes opposite of the direction indicated on the jog button, then the motor is wired backwards. Change the motor wiring. 2. Check the motor cabling, paying particular attention to the ground connections. Replace the cable if it is damaged or repair the motor connections. 3. Jog the motor for awhile at the maximum rate using the fast jog buttons (check the fast jog rate in the motor jog parameters screen to make sure that it is set to be equal to the maximum motor rate). If the motor seems to jump around rather than accelerate and decelerate smoothly, then you are probably facing an encoder error. Swap the motor with one from another axis and see if the error follows the motor. If it stays with the axis, replace the MPU11. If it follows the motor, replace the motor cable. If the problem still persists, replace both the motor and encoder.

Error	Message	Cause & Effect	Action
411	_ axis full power without motion	90% Power (a PID Output greater than 115) is applied to any axis and no motion greater than 0.0005 inches is detected for more than the time specified in Parameter 61 (the default is .5 sec.). All axis motion is stopped and the CNC program is aborted. The probable causes of this error are: 1. One of the axes is up against a physical stop. 2. The servo drive has shut down due to a limit switch input. 3. The Z-home switch is the same as the Z+ limit switch.	1. If the axis has run into a physical stop, use slow jog mode to move the axis away from the stop. Determine and set the software travel limits to stop the machine before it runs into the physical stop. 2. If the axis is not on a physical stop, check for a tripped limit switch. If one is tripped, then the software is commanding a move into the switch but the hardware is shutting the move down. Go to the motor setup screen and enter the limit switch input number, if applicable. 3. Make sure that the switch's input signal is not unstable or noisy. If it is, then replace the switch. If the problem persists, it may be necessary to create separate home and limit switch inputs. Use slow jog to move opposite of the direction causing the error and clear all of the limit switches. Jog toward the direction causing the error, and if no motion occurs, then this indicates that a servo drive failure has occurred.
412	_ axis encoder differential error	An error condition was detected in the differential signal levels for this axis encoder. This may indicate a loose/severed encoder cable or a bad encoder. This will stop all motion and cancel the job.	Reconnect/replace the encoder or encoder cable.
417	Abnormal end of job	Job was ended without reason.	
418	Search Line or Block not found	The requested search input data was not found in the loaded CNC file.	Type in the correct search input data or load the correct job.
419	Search line in embedded subprogram	The requested search line is found, but is part of an embedded/extracted subprogram.	Use another line number.
420	_ axis motor overheating	The CNC software estimates that one or more motors have reached the warning temperature (set in Parameter 29). Either the motor/s are overheating or the temperature file is corrupted. The job will be cancelled.	Contact dealer. Determine what is causing the motor/s to overheat, or delete the <code>cnc.tem</code> file and reboot.

Error	Message	Cause & Effect	Action
421	Motor(s) too hot: job canceled	The CNC software estimates that one or more motors have reached the limit temperature (set in Parameter 30). The job will not be able to run until the motor/s cools down.	Contact dealer. Determine what is causing the motor/s to overheat or delete the <code>cnc.tem</code> file and reboot.
422	Check Jog Panel cable	Jog panel failure or loose cable.	Reconnect the jog panel cable.
428	Check MPG cable	MPG failure, loose cable, or it is turned off.	Reconnect the MPG cable and turn the axis selector knob to an axis.
434	_ idling too high: Releasing power	The axis is not moving and no job is running, but the axis has stopped against some abnormal resistance. Power is released to the motors.	Run Autotune to adjust the motor settings.
435	_ axis runaway: Check motor wiring	Motor was in a runaway fault condition. Power to the motor will be automatically shut off.	Check the motor wiring.
436	Servo drive shutdown	This error message is produced by the hardware detection of a physical error. The servo drive hardware generates this error message if it detects either an overcurrent or overvoltage condition. The particular hardware condition detected is shown with the servo drive's LEDs. Once the servo drive detects this error condition, it stops all motion and removes power to the motors. The servo drive indicates the presence of this condition to the CNC software via the servo drive fault input to the PLC.	On DC systems, check the status of the servo drive's LEDs and check fibers 4&5. If this message is displayed on an AC system, check that P178 bit 4 is set.
437	Servo power removed	An axis was moving more than 300 RPM while the power was supposed to be off. 1. The motor may be wired backwards. 2. There may be a shorted servo drive. 3. The axis motion was canceled, but the motor continued to move due to inertia. This is probably caused by an unbalanced axis. Power to the motors is released.	Check the motor wiring, servo drive, or look at the Kg value in the PID and make sure it is not above +/-5.
438	Spindle slave position error	The slaved axis moved too far in the wrong direction during a spindle-slaved move (such as in rigid tapping). The job is cancelled.	Check Parameter 34 for an incorrect plus/minus sign in front of the encoder counts.

Error	Message	Cause & Effect	Action
439	_ axis servo drive data output error	Logic power failure or loss of communication from the drive to the MPU11.	Is the logic LED on? Check the fiber optic cables to the drive. For SD1 drives, make sure that the bus cables are shielded and as short as possible. Power the unit down and check the drive connections.
441	_ axis overvoltage	The input power has gone higher than 340VDC. The control board will shut down the drive and remove power. The motor brake will engage for five seconds in this condition.	Check to make sure that the input voltage is below 340VDC. If not, the incoming VAC needs to be lowered.
442	_ axis undervoltage	The drive's input power is less than 80 VDC.	Check the supply voltage.
443	_ axis commutation encoder bad	The control detected an invalid commutation zone value	Perform a motor Move Sync in the Drive Menu. A Zero (0) or Seven (7) is an invalid zone. Check for: 1. A wiring problem in the encoder cable or motor end cap (such as broken encoder wires). 2. The encoder cable shield being connected at the motor end, when it should not be. 3. A bad encoder. 4. Motor power cable shields not being connected properly. 5. A drive that is not grounded properly.
444	_ axis overtemperature detected	The drive overtemp sensor was tripped. The motor will not be given power.	Either the drive is being run at overcapacity, the cooling fan is not functioning, or the cooling fan's airflow is blocked.
445	_ axis overcurrent detected	Overcurrent is detected on an axis. The motor will not be given power.	Try to jog the axis. The drive will reset the current limit and try to move the motor. If the error comes back, check for a short in the motor output.
446	_ axis servo drive data input failure	Communication Checksum error. The motor will not be given power.	Check the fiber optic cables. Verify continuity between the drive chassis, ground strip, and Earth ground.
447	_ axis (#) bad index pulse detected	Noise was picked up by an encoder cable or misaligned encoder. The motor will not be given power.	Remove the noise or align the encoder.
449	Manual movement detected in restricted area	Unexpected manual movement of the axis was detected when Z-axis summing is active.	Physically lock the Z-axis manual quill.

Error	Message	Cause & Effect	Action
450	Voltage brake applied	Overvoltage condition was detected. Electronic braking was applied by offloading excess voltage to the dropping resistors.	Usually this error condition is innocuous, even if this message occurs every once in a while in a job. However, if this message occurs in a continuous stream, contact your dealer.
451	Current brake applied	Overcurrent spike was detected on the drive.	Usually this error condition is innocuous, even if this message occurs every once in a while in a job. However, if this message occurs too often, it may mean you need a drive with a higher current. But, if this message appears in a continuous stream, then something is seriously wrong. You should hit the E-Stop to cut power to the drive and then contact your dealer.
452	PC Receive Data Error	A fatal communication error occurred between the MPU and PC. This error was detected on the PC side.	Restart the software to clear the error. If this error occurs often, there may be an issue with the network configuration or Ethernet cable.
453	CPU Receive Data Error	A fatal communication error occurred between the MPU and PC. The error was detected on the MPU11 side.	Restart the software to clear the error. If this error occurs often, there may be an issue with the network configuration or Ethernet cable.
453	Jogging while probe detected	The probe was in a tripped state while a jog key was being pressed.	
454	axis scale encoder differential error	An error condition was detected in the differential signal levels for this axis scale encoder. This may indicate a loose/severed encoder cable or a bad encoder. This will stop all motion and cancel the job.	Reconnect/replace the scale encoder or scale encoder cable.
455	axis encoder quadrature error	The axis encoder skipped a transition state in its count-up/count-down sequence. This may indicate a loose/severed encoder cable or a bad encoder. This will stop all motion and cancel the job.	Reconnect/replace the encoder or encoder cable.
456	axis scale encoder quadrature error	The scale encoder skipped a transition state in its count-up/count-down sequence. This may indicate a loose/severed encoder cable or a bad encoder. This will stop all motion and cancel the job.	Reconnect/replace the scale encoder or scale encoder cable.

Error	Message	Cause & Effect	Action
457	Unable to find home	A commanded move was seeking either an index pulse or a hard stop, but neither was found.	Reconnect/replace the encoder or encoder cable if the move was seeking an index pulse. Check that the hard stop was not broken off or overrun.
459	TT1 or Probe is not connected	A Tool Measure operation was aborted because the required TT1 or Probe is not connected.	Check the TT1 or Probe wiring and plug.
460	TT1 and Probe are both connected	A Tool Measure operation was aborted because both a TT1 and Probe were connected.	Make sure that the TT1 and Probe are not plugged in at the same time. Also check the wiring.
461	Spindle axis is not set	An operation aborted because the spindle axis parameter (35) has an incorrect value.	Contact dealer.
462	Triangular Rotary Axis Out of Range	A triangular rotary axis (on a tilt table or articulated head machine) is at a position which is out of range. for angular calculation.	Contact dealer.
470	brake wattage exceeded	The brake wattage was exceeded on the indicated ACDC drive.	Contact dealer.
487	Invalid tilt table parameters	One or more values in the tilt table configuration are incorrect.	Contact dealer.
490	Reset Initiated, Press Reset to clear.	The reset state has been set and needs cleared by pressing Reset	Press Reset to clear state.

16.5 CNC Syntax Errors

Error	Message	Cause & Effect	Action
501	Invalid character on line NNNNN	Invalid character on CNC line. Job cancelled.	Remove character from program.
502	Invalid G-code on line NNNNN	Invalid G-code encountered on CNC line. Job cancelled.	Correct the invalid G-code.
503	Invalid M-function on line NNNNN	Invalid M-function encountered on CNC line. Job cancelled.	Correct the invalid M-code.
504	Invalid parameter on line NNNNN	Invalid or missing number after letter. Job cancelled.	Correct the program.
505	Invalid value on line NNNNN	Value out of range (T, H, D). Job cancelled.	Correct the program.
506	Only 1 M-code per line	More than one M-code appears on the line. Job cancelled.	Move 2 nd M-code to next line.
507	No closing quote	The closing quotation mark (") is missing. Job cancelled.	Add the missing quotation mark.
508	Macro nesting too deep	The macro nesting limit was exceeded on an attempt to invoke a subroutine. Job cancelled.	Create a second program.
509	Option not available	An attempt was made to access a locked software option. Job cancelled.	Contact Dealer.
510	Too many macro arg's	Too many arguments were given in a G65 macro. Job cancelled.	Correct the number of arguments.

Error	Message	Cause & Effect	Action
511	Missing parameter	A parameter is required or expected but not found. Job cancelled.	Correct the program.
513	Expected “=”	There was an error in the expression to the left of “=”, a missing “=”, or an orphaned parameter. Job cancelled.	Correct the equation.
514	Empty expression	The expression contains no operands. Job cancelled.	Correct the expression.
515	Syntax error in expression	An illegal character was in the number, variable, or function. Job cancelled.	Correct the program.
516	Unmatched bracket (parenthesis)	Brackets or parentheses are paired improperly or misplaced. Job cancelled.	Correct the program.
517	Evaluation stack overflow	Brackets or parentheses are nested too deeply. Job cancelled.	Correct the program.
518	Undefined variable	The variable name does not exist. Job cancelled.	Correct the program.
519	Too many variables	The space allotted for user-defined variables has been exceeded. Job cancelled.	Correct the program.
520	Invalid variable name	The variable name contains an illegal character. Job cancelled.	Correct the program.
521	Divide by zero	There was an attempt to divide by zero. Job cancelled.	Correct the program.
522	Domain error	An imaginary number would result (there was a square root of a negative number). Job cancelled.	Correct the program.
523	Invalid value in assignment	There was an attempt to assign an illegal value to a system variable. Job cancelled.	Correct the program.
524	Variable is read-only	There was an attempt to assign a value to a read-only system variable. Job cancelled.	Correct the program.
525	Missing P value	A P parameter was expected but was missing	Correct the program.
526	M22x Missing initial variable	M224 or M225 were not immediately followed by a #variable reference.	See M224 and G225
527	M22x initial variable parse error	M224 or M225 were immediately followed by an invalid #variable reference.	Correct the program.
528	M225 String variable not allowed	M225 was immediately followed by a string #variable (which is invalid). Only numeric variables are allowed here.	Correct the program.
529	M225 invalid variable	The #variable specified after the M225 was not valid, or not readable due to a machine error.	Correct the program.
530	M224 invalid variable	The #variable specified after the M224 was read-only, or not writeable due to a machine error.	Correct the program.
531	M22x missing initial quote	The beginning of the quoted (") format string was not found or was in the wrong place on the G-code line.	See M200 , M223 , M224 , or M225

Error	Message	Cause & Effect	Action
532	M22x missing end quote	The format string did not end with a quote (").	See M200 , M223 , M224 , or M225
533	M22x embedded quote not allowed	The format string contained a quote (") in the middle of it.	See M200 , M223 , M224 , or M225
534	M22x character limit exceeded	The format string was too long.	Correct the program.
535	M22x invalid format string	The format string contained invalid format codes.	Correct the program.
536	M22x missing format specifier	The format code was missing its specifier.	Correct the program.
537	M22x Missing Argument	A format code was specified in the format string, but its corresponding #variable argument was missing.	Correct the program.
538	M22x argument parse error	A format code was specified in the format string, but its corresponding #variable argument had a syntax error	Correct the program.
539	M22x variable type mismatch	A string format code was specified in the format string, but its corresponding #variable argument was numeric OR a numeric format code was specified in the format string, but its corresponding #variable argument was a string	Correct the program.
540	M22x variable cannot be read	A format code was specified in the format string, but its corresponding #variable argument was invalid OR there was a machine error when accessing it.	Correct the program.
542	M22x character limit exceeded	The resulting formatted string after all of the format codes were processed was too long.	Correct the program.
543	Missing L parameter	An L-code was missing	Correct the program.
544	Too many axes	More than one axis was specified with M128 OR the Simultaneous Contouring feature is not enabled. Without the Simultaneous Contouring feature, a maximum of three axes are allowed per G-code line.	Specify fewer axes on the G-code line OR Contact Dealer for information about obtaining the Simultaneous Contouring feature.
545	Value out of range	A parse error occurred because the value was out of range.	Correct the value.
547	Move by counts not allowed	Cutter comp (G41/G42) was on when M128 was specified.	Issue G40 (Cutter compensation off) before issuing M128 .
548	String too long	A quoted string was too long (usually a file name was longer than its allowed limit).	Shorten the file name.
549	Line too long	A line in a G-/M-code program is too long (more than 1023 characters).	Shorten the program line.
550	Invalid L parameter	The value associated with the L-code is invalid.	Give the correct value.
551	Invalid R value	The value associated with the R-code is invalid.	Give the correct value.

Error	Message	Cause & Effect	Action
552	File encryption error	Error while parsing encrypted G-code file.	

16.6 Cutter Compensation Errors

Error	Message	Cause & Effect	Action
601	Error: no compensation in MDI	G41 or G42 was entered while in MDI. MDI is not canceled, but the cutter compensation does NOT go into effect. The remainder of the line is processed.	Do not use G41 or G42 in MDI.
603	Arc as first uncomp. move on line NNNNN	An arc was specified as the first move after the end of compensation (G40). Job cancelled.	The first move after G40 must be a linear move.
604	Plane must be XY on line NNNNN	Cutter compensation was started with the YZ or ZX plane selected. Job cancelled.	Remove cutter compensation for YZ or ZX plane moves. This option is not available.
605	Canned cycle not allowed on line NNNNN	A canned cycle was attempted during compensation. Job cancelled.	Do not use cutter compensation with canned cycles.
606	G53 not allowed on line NNNNN	A G53 was attempted during compensation. Job cancelled.	Choose a different work coordinate.
607	Set home not allowed on line NNNNN	A M26 was attempted during compensation. Job cancelled.	Do not use M26 with cutter compensation
608	Ref. point move not allowed on line NNNNN	A G28, G29, or G30 was attempted during compensation. Job cancelled.	Do not use return points with cutter compensation.

16.7 Parameter Setting Errors

Error	Message	Cause & Effect	Action
701	G10 error: no R-value on line NNNNN	G10 used with no R-value. Job cancelled.	Input an R-value.
702	G10 error: invalid D on line NNNNN	Job cancelled (D0 cannot be set; it is always zero).	Change D to a valid value.
703	G10 error: invalid H on line NNNNN	G10 H0 Rxx was specified. Job canceled (H0 cannot be set; it is always zero).	Change H to a valid value.
704	G10 error: invalid P on line NNNNN	A G10 was used with an unknown P value. Job cancelled.	Change P to a valid value.
705	G10 error: No D, H, or P on line NNNNN	A G10 was used without a D, H, or P to assign a value. Job cancelled.	Add a valid D, H, or P value.

16.8 Canned Cycle Errors

Error	Message	Cause & Effect	Action
801	Error: No R-point on line NNNNN	No R-value was specified. Job cancelled.	Add an R-point.

Error	Message	Cause & Effect	Action
802	Error: Q = 0 on line NNNNN	A Q-value of 0 was specified (Q is used for G73 and G83 only). Job cancelled.	Insert a non-zero Q-value.
803	Error: No Z-point on line NNNNN	No Z-value was specified for the canned cycle. Job cancelled.	Add a Z-value.
804	Error: Ggg invalid on line NNNNN (gg = 76, 86, 87, 88)	An unimplemented canned cycle was requested. Job cancelled.	Change to a valid G-code.
805	Error: No Q-value on line NNNNN	A Q-value was not specified for G73 or G83. Job cancelled.	Insert a Q-value.
806	Error: No P-value on line NNNNN	A P-value (dwell time) was not specified for G82 or G89. Job cancelled.	Add a P-value.
807	Error: Cannot execute G__ when axis B is rotated	On an Articulated Head machine with the TWCS feature enabled, a non-compound canned cycle (such as G73, G74, G76, G80, G81, G82, G83, G84, G85, or G89) was issued on a WCS that was set to TWCS=No while the spindle head was tilted (i.e. rotary B axis was not at 0). Job cancelled.	Either move B to 0 or issue the compound canned cycle version of the erroneous G-code, such as G173, G174, G176, G181, G182, G183, G184, G185, or G189.

16.9 Miscellaneous Errors/Messages

Error	Message	Cause & Effect	Action
901	Ref. point invalid on line NNNNN	A G30 with an invalid P-value (must be 1 or 2) was issued. Job cancelled.	Change the P-value to 1 or 2.
902	No prior G28 or G30 on line NNNNN	A G29 with no preceding G28 or G30 was issued.	Add a G29 or G30.
903	Warning: No coordinates for G92 on line NNNNN	G92 with no axis coordinates to set. Remainder of line processed; job continues.	Add coordinates.
905	Warning: 0 radius arc on line NNNNN	Arc move was specified with a zero radius. Move is done as a linear move; job continues.	Specify a radius.
906	Warning: unknown arc on line NNNNN	Position of arc move could not be determined from parameters (e.g. G91 G2 X0 Y0 R1). Move is done as a linear move; job continues.	Correct program.
907	_ axis travel exceeded on line NNNNN	Software travel limit would be exceeded by the requested move. Job cancelled.	Check program, part zero, or tool offset.
909	Program too long: job canceled	Attempt to run a job over 1MB in length, without the unlimited program size option. Job cancelled.	Contact Dealer or break up program.
910	No subroutines in MDI	Specified O9100–O9999 in MDI, which would begin an embedded subprogram. MDI cancelled.	Insert subroutines into MDI.

Error	Message	Cause & Effect	Action
911	Illegal recursion	Attempt to execute a subprogram or macro that calls itself, either directly or indirectly. Job cancelled.	Call correct subprogram.
913	Could not open file filename.ext	Attempt to call a subprogram or macro, but the subprogram file does not exist. Job cancelled.	Make sure file name is correct and is in the ncfiles directory.
915	DSP window retry sN fN rN	DSP window checking failed, move will be repeated unless the maximum retries have been reached, s = number of successes, f = number of failures, r = number of times the maximum retry value has been reached.	
916	Unexpected probe contact	Probe tripped when a cycle did not expect contact.	
917	Invalid tilt lookup table	The tilt lookup table file (tilt.tab) has an invalid format or was not found.	
918	Probe unable to detect surface	Probe travelled maximum distance without contact, dsp window checking failed, or probe repeatability failed.	
919	DSP window failed maximum retries	DSP probe reached the maximum retry limit without a successful window.	
920	Unable to clear obstacle	Probing cycle failed to clear an obstacle.	
921	Unable to determine corner	Probing cycle failed to find corner (inside and outside corner).	
922	Out of memory	Problem allocating memory.	
923	Error: Z-home not set	Z-home is not set.	Set Z-home
924	File read error	Problem reading the job file. This error occurs if the file was opened successfully, but there was an error while reading the file.	
925	Error reading job file	Same as above, but at a different place in the code.	
926	Failed to locate job continuation position	Job continuation from the Run Menu failed.	Do a Run/Search
927	Too many subprogram calls	Nesting level of subprograms is too high. I.e. a subprogram calls another subprogram which calls another subprogram, which calls another subprogram, etc. . .	Reduce the nesting level of subprograms.
928	Error Loading Log Configuration file. . . Using defaults	There was an error while loading the log configuration file. Default settings will be used.	
929	Log Level set to ____	The logging level parameter (140) has been changed.	No action required.
930	Log Level Configuration file not found. . . Creating new configuration.	The log level configuration file was not found. A default file will be created.	No action required.
931	Error during transformed move to home	A transformed moved to home (M25) command attempted to move below the G28 Z-position.	

Error	Message	Cause & Effect	Action
932	Error during Tool Check	A general error condition occurred when the Tool Check key was pressed.	
933	Log file initialized	There was an error in trimming the log file or it did not exist, so a new log file has been created.	
934	Warning: Excess precision truncated	A CNC program is using axis positioning precision greater than what is displayed, and therefore the actual commanded positions are truncated. This happens when the Simultaneous Contouring feature was not enabled. This feature must be enabled for the extra precision to be acknowledged.	Contact Dealer for information about obtaining the Simultaneous Contouring feature.
935	_ axis (#) scale disabled	A scale is enabled for this axis but compensation was disabled. Scale compensation is disabled at initial power up, configuration changes, and during homing moves.	Home the machine.
935	Probe failed reset retries	Probe failed to reset after three tries. The probing operation may have been started too close to the surface.	Move the probe further away from surface and do the probing operation again. If this continues to fail persistently, call dealer.
936	_ axis (#) scale enabled	A scale is enabled for this axis and compensation was enabled. This happens after homing the axis.	Not Applicable
936	Probe failed to reset	Probe failed to reset. The probing operation may have been started too close to the surface.	Move probe further away from surface and do probing operation again. If this continues to fail persistently, call dealer.
944	MPU requested resend	The MPU requested a resend.	Status Message
945	PC requested resend	The PC requested a resend.	Status Message
946	PC resending	The PC is resending.	Status Message
947	PC received data out of order	The PC needed to reorder data received from the MPU.	Status Message
948	PC packet error	The PC received bad data from the MPU and will try to recover by requesting a resend.	Status Message
949	Drive map does not match hardware	One or more of the drive mapping parameters 300–307 is misconfigured.	Contact Dealer.

16.10 Scaling/Mirroring Errors

Error	Message	Cause & Effect	Action
1001	Invalid scaling parameter on line NNNNN	Invalid parameter specified (I, J, K, P). Job cancelled.	Remove or change invalid parameter.
1002	Invalid scaling center on line NNNNN	Invalid parameter specified (X, Y, Z). Job cancelled.	Remove or change invalid parameter.

Error	Message	Cause & Effect	Action
1003	G-code not allowed when scaling on line NNNNN	G28/G29/G30/G92 is not allowed when scaling or mirroring is turned on. Job cancelled.	Move G-code to appropriate line.
1004	Turn scaling off before rescaling	Tried to rescale while scaling is turned on. Job cancelled.	Turn scaling off, then rescale.
1005	Cannot scale arcs with different scale factors	Scaling factors of the arc axes are different. Job cancelled.	Correct scaling factors, or separate scaling operations.
1100–1199	Custom messages defined in <code>cncxmsg.txt</code> . Please contact your dealer if you have any questions regarding a particular message. This style of message should be replaced with the <code>plcmmsg.txt</code> format on MPU11 systems.		

16.11 Configuration Modification Messages

Error	Message	Cause & Effect
111	___ modified: ___ → ___	An axis configuration parameter was modified.
444	___ modified: ___ → ___	A servo drive configuration parameter was modified.
555	___ modified: ___ → ___	A PID configuration parameter was modified.
556	Axis converted: → ___	A PID configuration parameter was converted.
777	___ modified: ___ → ___	An axis configuration parameter was modified.
888	G30 Z modified: → ___	Z-coordinate of Secondary Reference Point was modified.
999	Parm # modified: → ___	A machine parameter was modified.